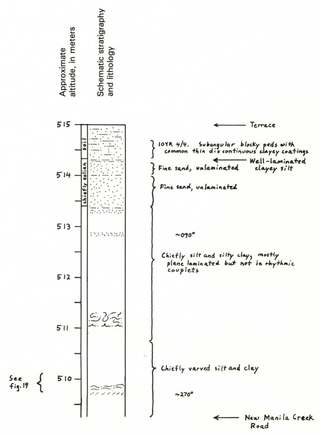
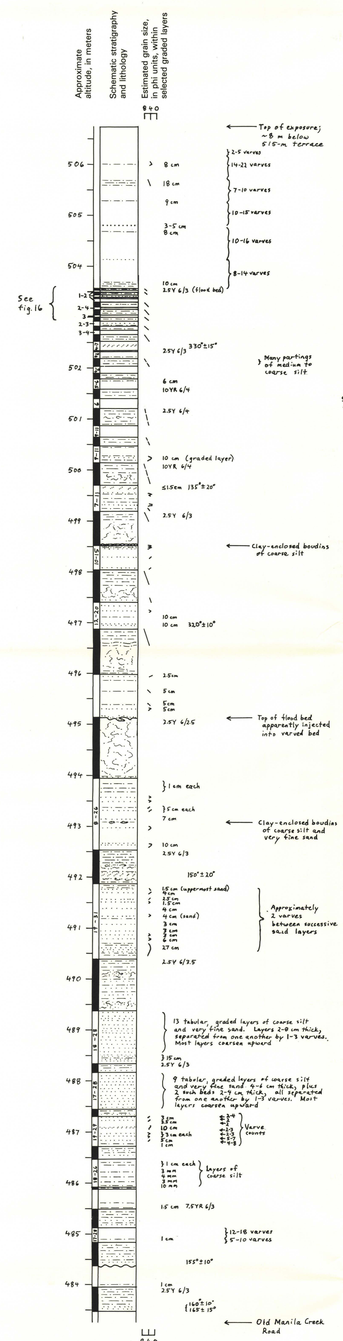
DEPARTMENT OF THE INTERIOR
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BULLETIN 1661
PLATE 3

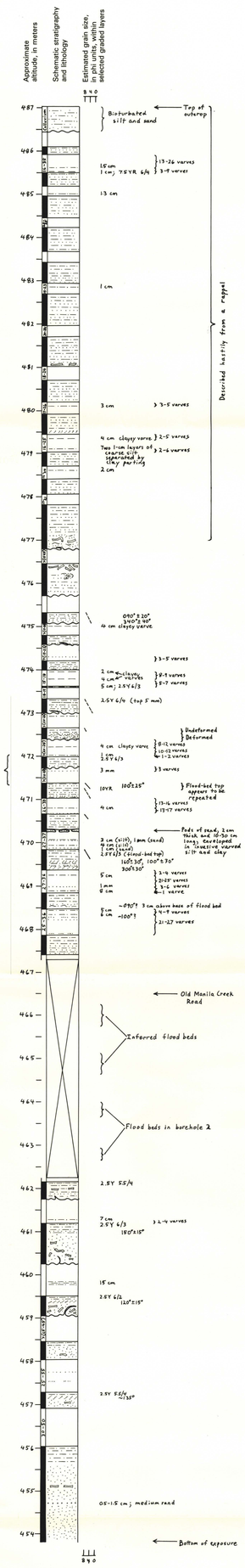
Last Chance locality



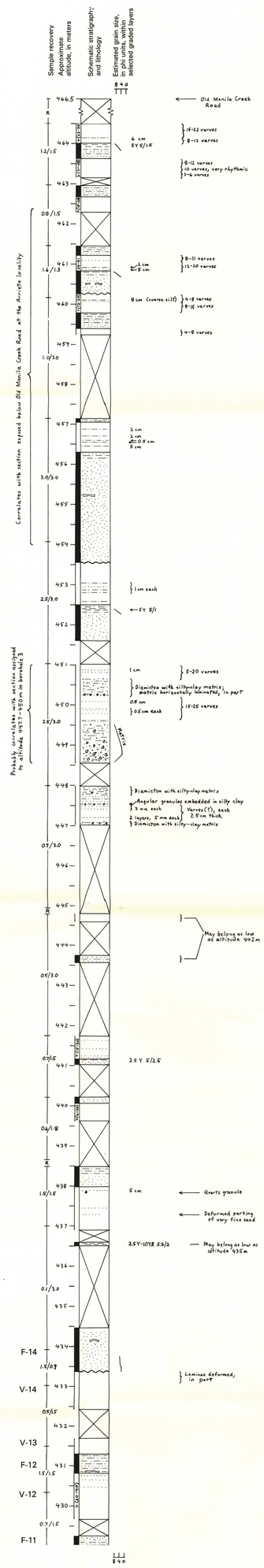
Switchback locality



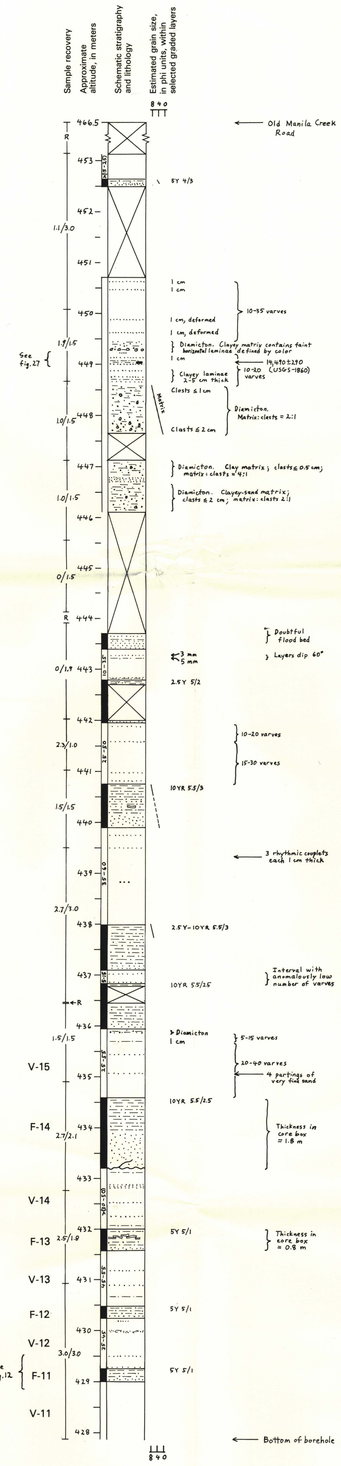
Arrieta locality



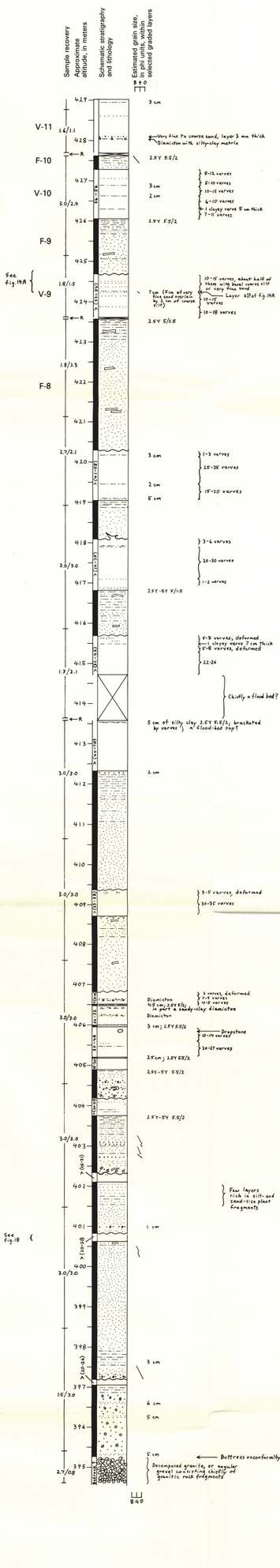
Borehole 2 (upper part)



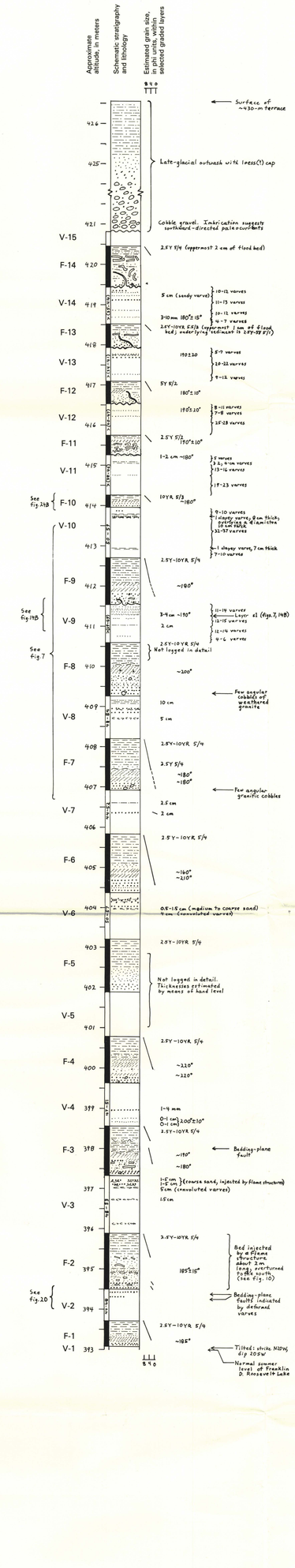
Borehole 3



Borehole 2 (lower part)



French Johns locality (main part)



EXPLANATION

(see plate 2A for map location, plate 2B for composite stratigraphic columns)

Generalized stratigraphy

VARVED INTERVAL—Numbers in narrow column give stingy-to-generous range of varve counts
INTERVAL COMPLETE OR NEARLY SO
INTERVAL INCOMPLETE. Varve count is limiting minimum owing to erosion of the top of the interval (general case), non-recovery of core (borehole 2, 462.5 m), or structural thinning (northernmost part of French Johns locality, 395.3 m)
FLOOD BED
OTHER KIND OF MATERIAL—As labeled

Lithologic columns

GRAIN SIZE—In varved intervals, shown only for strata with notable characteristics
CLAY
SILT
VERY FINE SAND AND FINE SAND
MEDIUM SAND AND COARSE SAND
GRANULES AND PEBBLES. Chiefly angular
COBBLES AND BOULDERS. Chiefly rounded

SEDIMENTARY STRUCTURES

RIPPLE-DRIFT CROSS LAMINATION—Shown chiefly where sets thicker than 5 cm
LENSES OF SAND
POCKETS OF SAND
CONVOLUTED LAMINATION
FLAME STRUCTURES. Height and shape approximately to scale

CONTACTS BETWEEN VARVED INTERVALS AND FLOOD BEDS

CONFORMABLE OR NEARLY SO
DISCONFORMABLE. Some disconformities accompanied by flame structures

MISCELLANEOUS CONSTITUENTS

RIP-UP CLASTS. Composed chiefly of varves
WOOD. Altitude 449 m in borehole 3
PROBABLE BEDROCK. Bottom of borehole 2
LITHOLOGY UNKNOWN—Interval of unrecovered core (boreholes) or of concealed exposure (Arrieta locality)

Miscellaneous lithologic features (right side of lithologic column)

GRAIN-SIZE TRENDS IN SELECTED GRADED LAYERS—Grain size (see scale at top and bottom of column) estimated in the field. Line length denotes thickness of graded layer. Dashes indicated interrupted grading
UPWARD FINING
UPWARD COARSENING
UPWARD COARSENING SUCCEEDED BY UPWARD FINING
1 cm — THICKNESS OF LAYER WITH NOTABLE CHARACTERISTICS—Shown chiefly in varved intervals
2.5Y 6/4 — COLOR IN MUNSELL SYSTEM—Shown only for the uppermost parts of flood beds, and for a possible flood-bed fragment at altitude 413.5 m in borehole 2. Names of principal colors: gray (5Y 5/1), grayish brown (2.5Y 5/2), light olive brown (2.5Y 5/4), light yellowish brown (2.5Y 6/4), yellowish brown (10YR 5/4), dark yellowish brown (10YR 6/4), light brown (7.5Y 6/4)
220°±15° ~210° — PALEOCURRENT DIRECTION INDICATED BY CROSS LAMINATION—Error (±) shown where estimated

Sample recovery (left side of lithologic column; boreholes only)

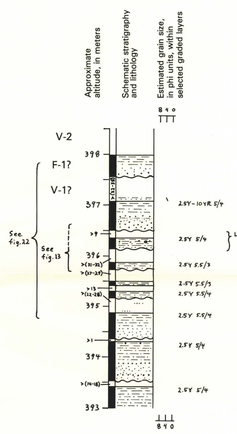
DRIVE (INTERVAL OF ATTEMPTED RECOVERY OF A CORE)—Denominator gives length of drive and numerator gives length of recovered core, both in meters. Kinds of recovery, with examples from borehole 3:
2.5/3.0 — FULL—434.9-436.4 m
2.7/3.0 — PARTIAL—Exact altitude for recovered material is commonly unknown. 436.5-439.5 m
0/1.9 R — NONE. Recovery attempted—442.0-443.9 m. Reamed interval—Recovery not attempted; borehole cleaned of previously unrecovered material. 436.5 m
EXCESSIVE
2.3/1.0 — Caused by partial recovery of material left in borehole from previous drive—441.0-442.0 m
2.7/2.1 — Cause unknown—Extra material assigned chiefly to flood-bed sand. 432.8-434.9 m

Control points for estimates of altitude in measured sections

FRENCH JOHNS LOCALITY—Summer-level shoreline of Franklin D. Roosevelt Lake, 393 m according to Keller 15-minute quadrangle (pl. 2A)
ARRIETA LOCALITY AND BOREHOLES—Surface of Old Manila Creek Road estimated at 466.5 m by interpolation from 1:2,400-scale topographic map with 1.5-m contour interval; map prepared for AMAX, Inc., by IntraSearch (Denver, Colorado) from 1:12,000-scale aerial photographs dated 6 July 1978 (IntraSearch project 79201)
SWITCHBACK LOCALITY—(1) Distinctively reddish clayey layer, correlated with and assumed to have the same 485.5-m altitude as a similar layer at the Arrieta locality (labelled RC in pl. 2B). (2) Top of exposure, estimated by hand levelling as 8 m below terrace shown at altitude 515 m on IntraSearch map
LAST CHANCE LOCALITY—Top of exposure, equivalent to a 515-m terrace on IntraSearch map

The stratigraphic columns and borehole data shown on this plate were photocopied from a tracing of the author's field notes and sketches.

French Johns locality (northernmost part)



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DETAILED STRATIGRAPHIC COLUMNS FOR GLACIOLACUSTRINE DEPOSITS IN THE MANILA CREEK AREA, WASHINGTON